Tutorial 8 report
EE5311 (Digital IC design)

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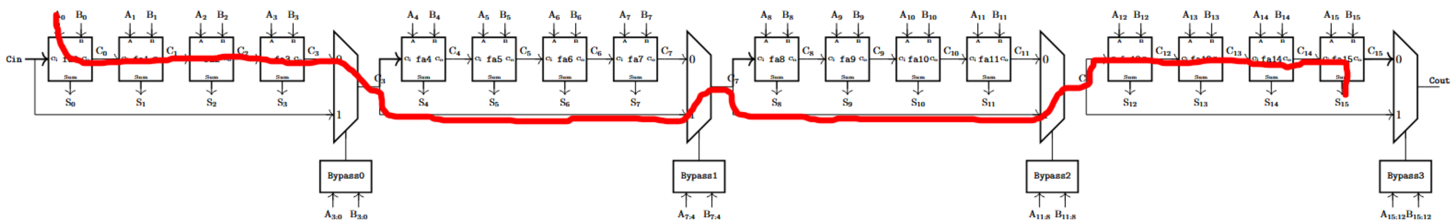
Question 1

Part a)

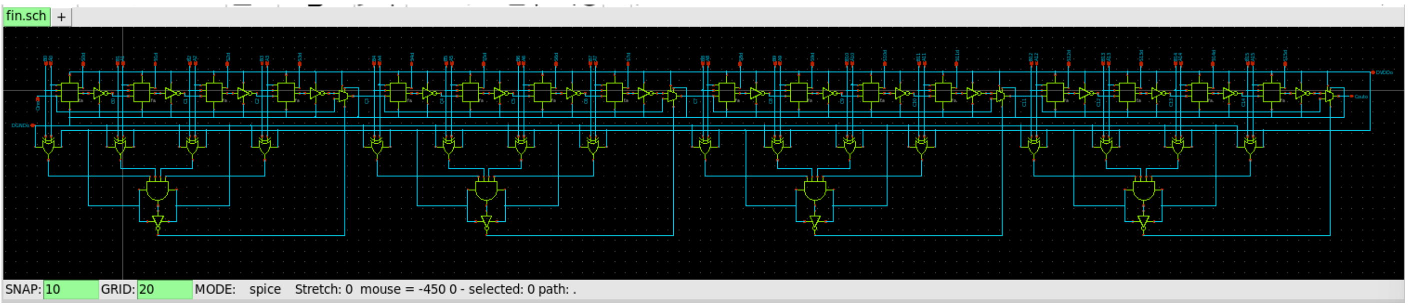
- The critical path, starting from the input, goes through all the 4 full adders in the 1st stage (where the 1st full adder generates/kills the carry and the other 3 adders propagate). The path then passes through port 1 of the MUXs at the end of the 2nd and 3rd stages. After this, it propagates through the 1st 3 full adders of the last stage and ends at the sum of the last full adder, S15.
- Traced critical path on the 16-bit carry look ahead adder –



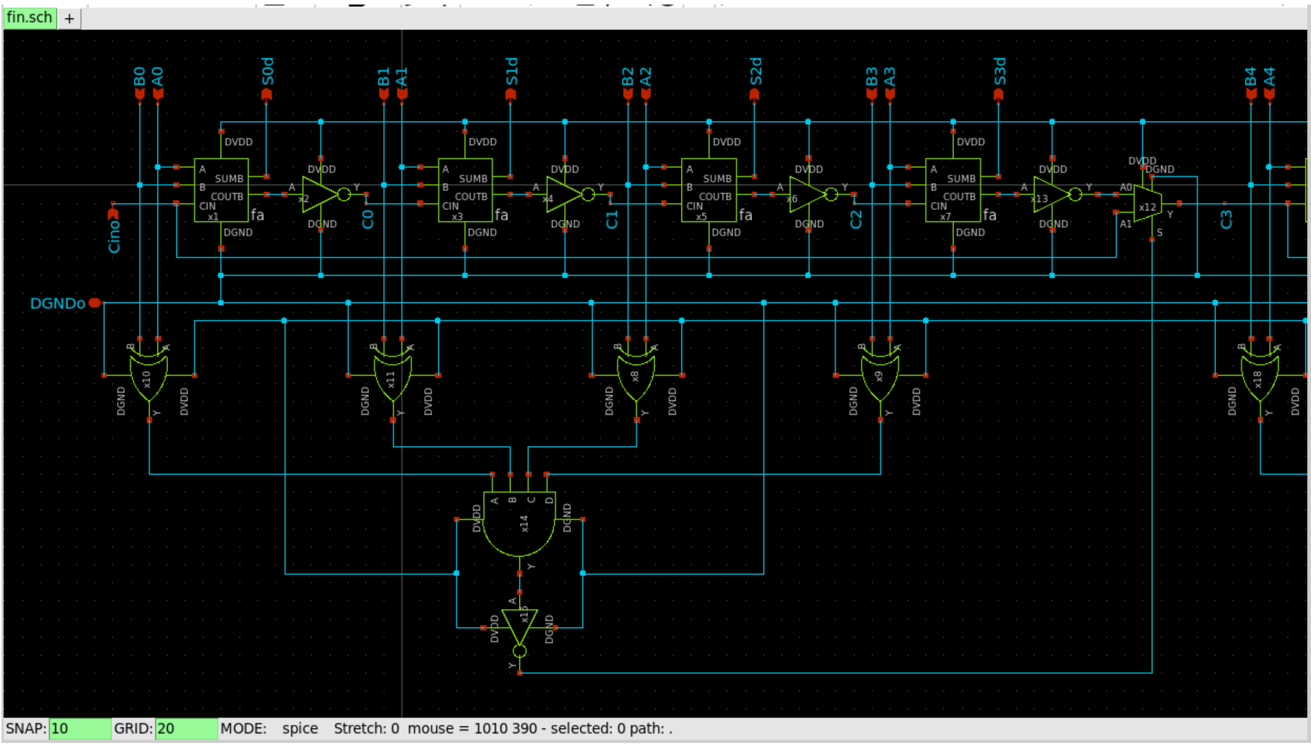
- An input combination that activates the critical path would be –
  A = 1111111111111111
  B = 0000000000000001
  Cin = 0
- The critical path delay would be –
  $t_{critical} = 4\ t_{carry} + 3\ t_{mux} + 3\ t_{carry} + t_{sum}$
  The first term 4 t_{carry} consists of 1 t_{carry} arising from the generate/kill delays of the 1st full adder and 3 t_{carry} from the propagation delay of the next 3 adders.
- Any other input combination, for example one that does not propagate through the 2nd and 3rd stages would be slower, since this means that a carry kill/generate would make the carry available immediately at that stage. This implies the availability of the output carry of the stage at the mux at at most 4 t_{carry} and that it would now have to travel through lesser MUXs to reach the final stage than if the entire stage had propagated.

Part b)

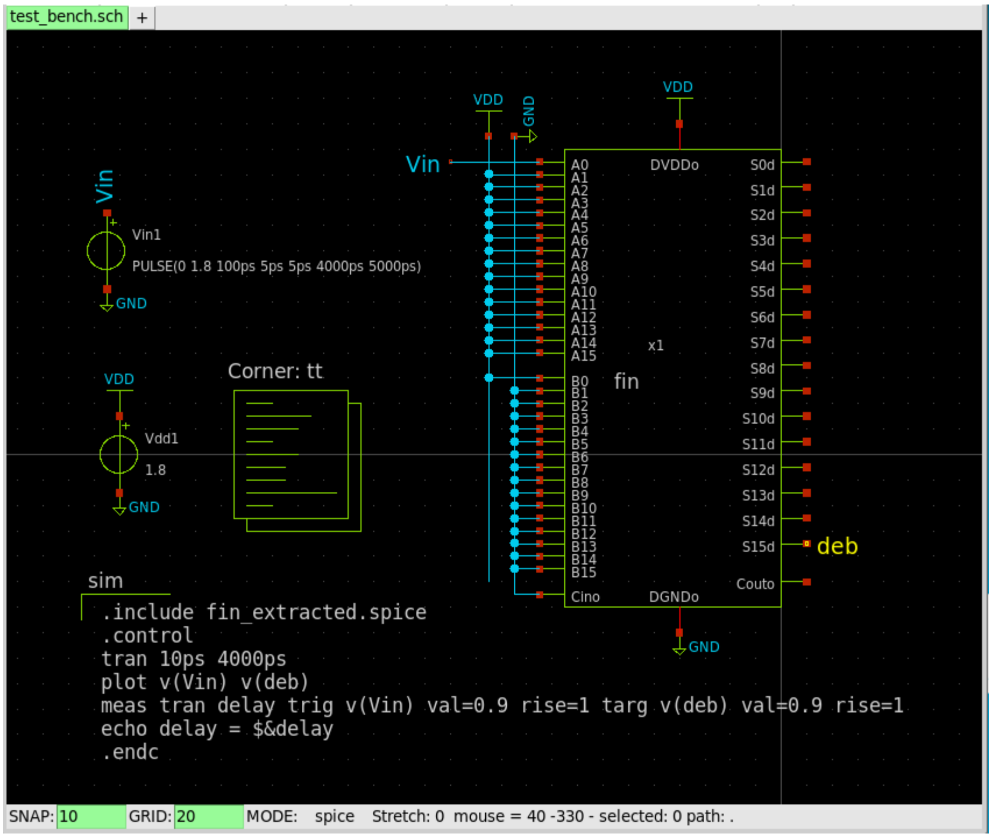
- The schematic used for the 16-bit carry look-ahead adder –



- A zoomed look at one of its 4 stages –

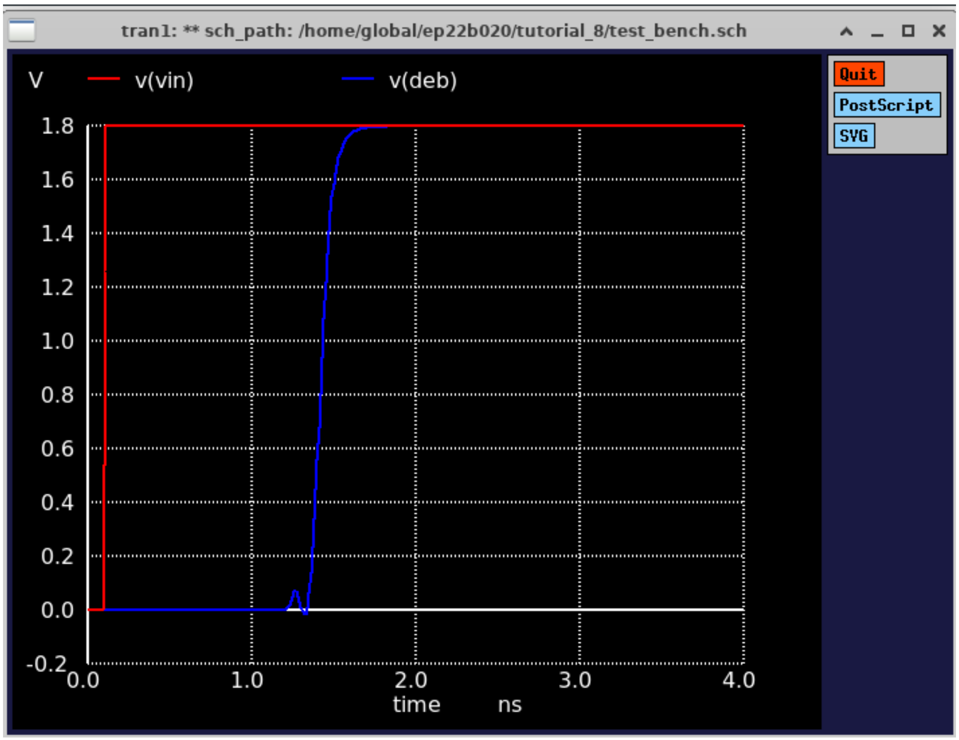


- xschem circuit and code used for measuring the critical delay –



- $t_{critical}$ = 1.322ns

```
 Reference value :  3.92740e-09
No. of Data Rows : 416
delay               =  1.322061e-09 targ=  1.424561e-09 trig=  1.025000e-10
delay = 1.32206E-09
ngspice 17 ->
```

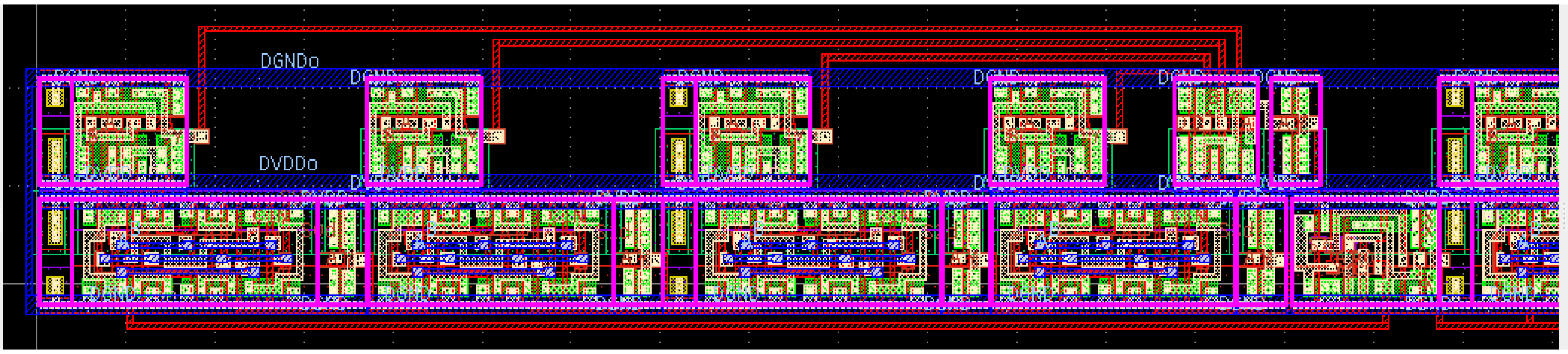


Question 2

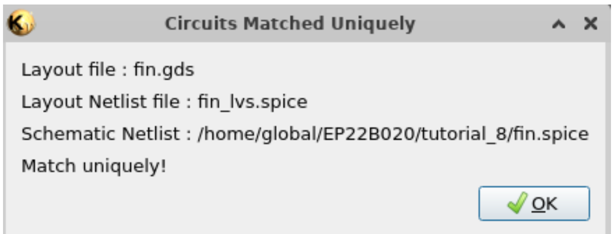
Part a)

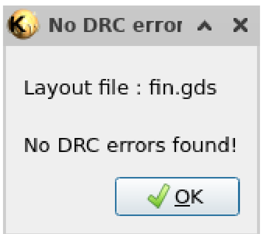
- Layout for the 16-bit carry look-ahead adder used –

- A zoomed look at one of its 4 stages –



- The layout passed DRC and LVS, ensuring its compatibility with the schematic –

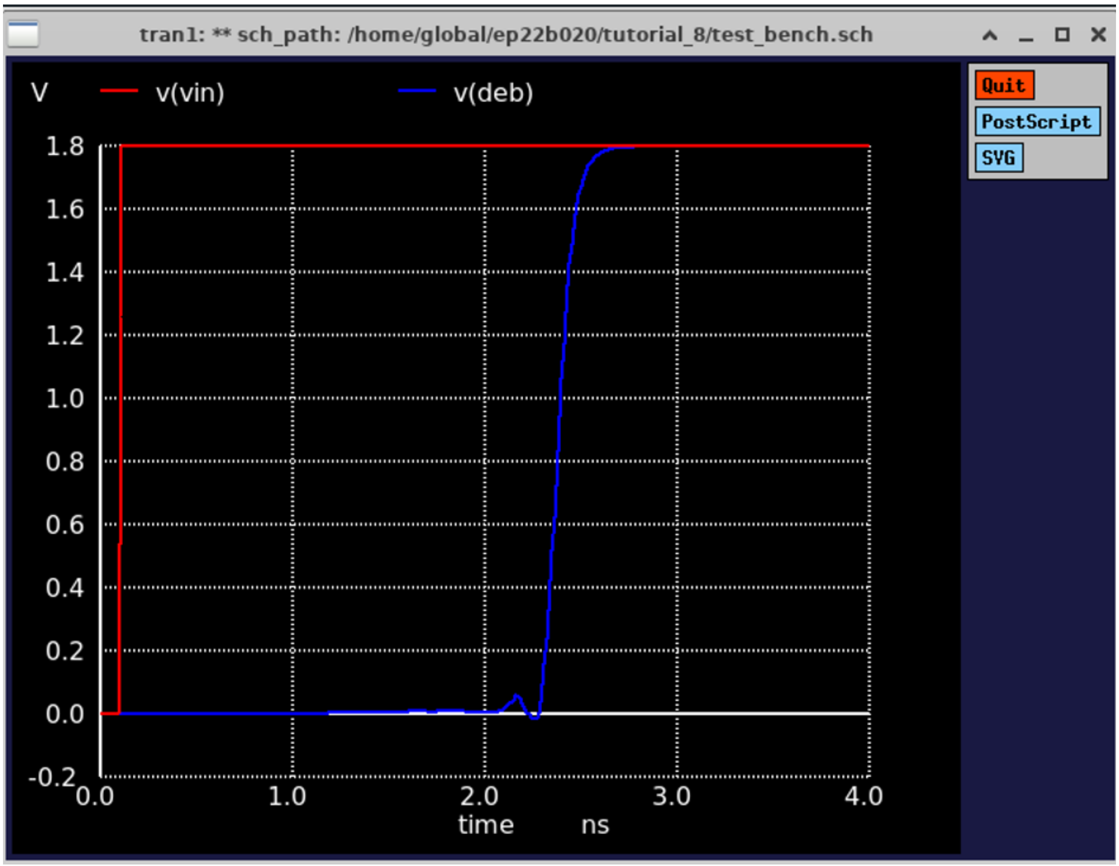




Part b)

- After extracting the layout parasitics into the schematic, $t_{critical}$ = 2.28ns

```
 Reference value :  4.00000e-09
No. of Data Rows : 416
delay               =  2.280308e-09 targ=  2.382808e-09 trig=  1.025000e-10
delay = 2.28031E-09
ngspice 17 ->
```



- We see that after including the parasitics arising from interconnects, the delay indeed increases.

Absolute path of layout

/home/EP22B020/ee5311/tutorial_8/fin.gds